Aviral Srivastava

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EDUCATION

Boston University, Boston, MA. MS, Computer Science. CGPA : 3.77/4 Sep 2019 - Dec 2020

Vellore Institute of Technology, Chennai, B.Tech, CS and Engg. CGPA : 3.56/4. Jul 2014 - May 2018

SKILLS

- Rust, Python, Scala, C++, Go, Java, PHP, SQL, Spark, Aurora, Mesos, Kubernetes, Docker, Vagrant, Argo, Airflow, Kafka, Flink, Iceberg, Debezium, Key-value DBs, AWS, GCP, Azure, On-Prem (Data Centers) Infrastructure

WORK EXPERIENCE

Affirm, Senior Software Engineer (L6), Lakehouse Infrastructure, NYC Mar 2025 - Present

- **Event-driven MySQL-to-Iceberg Replication:** designed and built it with Debezium, Kafka, Flink, and Iceberg for **2,000 tables**; validated data is now tagged in **10 minutes**, down from 60 on DMS.
  - Tagged data feeds downstream FinPlat pipelines, so Affirm can **buy and sell loans sooner**; the more reliable event-driven path also lowers **liquidity risk**.
  - Wrote ADRs for self-managed Debezium, **out-of-band validation**, and a validated stream writer using **GTIDs for ordering and transaction continuity**, plus the Kafka layout tech spec.
  - Merged six feature and correctness PRs into Debezium upstream; built **65+ failure-mode and load tests**, found 10 validation bugs, and shipped HA Kafka Connect.
- **Ledger on Lake:** directly responsible individual for shipping Affirm's ledger from MySQL to a lake-native architecture, saving **$XXM a year**.
  - Authored the read-cutover plan: schema-identical views over Iceberg, so **~138K reads a year** needed no query changes.
  - Root-caused **silent snapshot-chain breaks** that would fail incremental reads; fixed the retention policy in all three regions and added a daily check for new breaks.
  - Drove the **company-wide rollout**, aligning every team that reads the ledger on one plan and schedule for each regional cutover.
- Scale-tested **Iceberg compaction** on Flink for Black Friday peak load; profiled and optimized it.
- Led **incident response** for Ledger and Funding pipelines; facilitated the SEV1 retro for a Kafka CDC DNS outage (6 of 7 runbook changes adopted).
- Shepherded Data Storage and Services teams (Streaming, Online Storage, Batch) to establish SLAs, plan capacity, and agree on roles and responsibilities.

Pinecone, Senior Software Engineer, Platform, NYC Apr 2023 - Oct 2024

Fully managed vector database powering semantic search and RAG for AI applications.

- Single handedly expanded Pinecone's serverless offering to GCP from zero to GA, redesigning core services for cross-cloud compatibility, including auto-scaling for AlloyDB.
- Built Global Control Plane for the move to serverless with zero-downtime migration: request time from 10 seconds to about 1 second, operational costs down **10x**.
- Engineered **failover detection** and zero-downtime migration for **distributed database systems**; on-call for Serverless (v4) and Pod-based (v3) systems.
- Led cost engineering: moved CI/CD onto auto-scaling infrastructure, cutting spend by **$XXK/month**.

Twitter, Software Engineer II, Graph Storage(Platform/Real Time Storage), NYC Apr 2021 - Feb 2023

- We replaced the decade old Flock (Graph Storage service) with a new service (Flock v1.5) that will scale the write pipeline, improve data consistency, and reduce human ops. Led setting up v1.5 as a multi-data Center setup.
- Proposed, designed, and developed Backup Verification in **Scala**: a service to verify 300 TB of backups as they (backups) are used in v1.5 to serve live traffic.
- Performed oncall operations such as rebalancing shards, restoring servers during heatwave, modifying mirrorsets to cater fast product iterations, etc.
- Led the team for building a managed graph database offering for analytics purposes,graph machine learning, and applications. Reduced onboarding time from 1 week to minutes, for data teams at Twitter.

Atlan, Software Engineer(Data) May 2018 - Jul 2019

- Built and deployed Data Repository, version control for tabular data ('git diff', rollback) at **billion-row** scale on S3, RocksDB, Hive metastore and PrestoDB; scaled production releases by **300%**.
- Deployed UN SDG monitoring and Ujjwala dashboards on-premise, and integrated serverless JupyterHub used by Unilever, Mindshare and McKinsey.

Redhat, Software Engineering (Data) Intern, Internal Data-Hub team, Boston May 2020 - Dec 2020

- Re-partitioned the schema of SOS reports (**10 Trillion+ rows**, **100+ TB**) for high availability and low latency, using Argo, Spark and Kubernetes.

OTHER

- Open source: authored Debezium's official **TiDB connector**, released in Debezium 3.7.
- Projects: Zcash, Scalable UML, Polkadot-IPFS. Co-founder & CTO, Centify. Advisor, **Find Signal**.
- Srivastava, A., Patel, F., and Sivagami, M. *A Software Requirement Engineering Technique Using OOADA-RE And CSC For IOT Based Healthcare Applications*. IJSEA, 2018.